Ian Gershon Teixeira

Department of Mathematics, University of California San Diego
iteixeira@ucsd.edu · iangershonteixeira.github.io

Representation theory · algebraic coding theory · lattices · quantum information

Academic Appointments and Research Positions

University of California San Diego
Stefan E. Warschawski Visiting Assistant Professor of Mathematics — 2025–present

University of Maryland, College Park
MathQuantum Fellow, NSF Research Training Group in the Mathematics of Quantum Information — 2024–2025

University of Maryland / QuICS
Researcher, NSF Quantum Leap Challenge Institute for Robust Quantum Simulation — 2022–2023

Education

University of Maryland, College Park
Ph.D. in Applied Mathematics
Dissertation: *Quantum Codes, Transversal Gates, and Representation Theory*
Advisors: Jonathan Rosenberg (Mathematics) and Michael Gullans (Physics) — 2018–2025

Wesleyan University
B.A. in Mathematics, *Phi Beta Kappa* — 2013–2017

Publications

[1] E. Kubischta and **I. Teixeira**. "MacWilliams Identities for Intrinsic Quantum Codes." *IEEE Transactions on Information Theory*, accepted for publication (2026).

[2] E. Kubischta and **I. Teixeira**. "Nuclear-spin statistical weights from Young diagrams." *Journal of Chemical Physics* ***164****, 194120 (2026).*

[3] E. Kubischta and **I. Teixeira**. "Flexible fault-tolerant gate gadgets." *Physical Review A* ***113****, 022421 (2026).*

[4] E. Kubischta and **I. Teixeira**. "Classification of the subgroups of the two-qubit Clifford group." *Journal of Mathematical Physics* ***66****, 122203 (2025).*

[5] E. Kubischta and **I. Teixeira**. "Quantum codes and irreducible products of characters." *Designs, Codes and Cryptography* ***93****, 2919–2930 (2025).*

[6] E. Kubischta and **I. Teixeira**. "Permutation-invariant quantum codes with transversal generalized phase gates." *IEEE Transactions on Information Theory* ***71****(1), 485–498 (2025).*

[7] E. Kubischta and **I. Teixeira**. "Quantum codes from twisted unitary t-groups." *Physical Review Letters* ***133****, 030602 (2024).*

[8] E. Kubischta and **I. Teixeira**. "A family of quantum codes with exotic transversal gates." *Physical Review Letters* ***131****, 240601 (2023).*

Preprints

[P1] **I. Teixeira**. "From Steane to A_7: Quantum Codes from Invariant States." arXiv:2610.00865 (2026).

[P2] **I. Teixeira**. "A_2-frames, negacyclic glue, and extremal strongly 6-modular lattices." arXiv:2609.27148 (2026).

[P3] **I. Teixeira** and D. Meyer. "Deep Holes in the Clifford Hierarchy." arXiv:2608.08403 (2026).

[P4] E. Kubischta and **I. Teixeira**. "Time-Reversal Selection Rules for Quantum Error Correction." arXiv:2608.06304 (2026).

[P5] E. Kubischta and **I. Teixeira**. "Classical and Quantum MacWilliams Transforms as Spin Kinematics." arXiv:2607.27173 (2026).

[P6] E. Kubischta and **I. Teixeira**. "Minimal Permutation-Invariant Qudit Codes from Edge-Colorings of Complete Graphs." arXiv:2605.22439 (2026).

[P7] **I. Teixeira**. "Orthogonal Polynomials and the MacWilliams Transform for Permutation-Invariant Qudit Codes." arXiv:2605.15372 (2026).

[P8] E. Kubischta and **I. Teixeira**. "Intrinsic Quantum Codes." arXiv:2511.14840 (2025).

[P9] E. Kubischta, **I. Teixeira**, and J. Maxwell Silvester. "Quantum Weight Enumerators for Real Codes with X and Z Exactly Transversal." arXiv:2306.12526 (2023).

Selected Talks

- **Deep Holes in the Clifford Hierarchy**, RIT on Geometry and Physics, University of Maryland, September 17, 2026. *Invited seminar.*
- **Deep Holes in the Clifford Hierarchy**, QuEra Computing, September 10, 2026. *Invited seminar.*
- **Intrinsic Quantum Codes**, QASAR Seminar on Quantum Error Correction and Fault Tolerance, INRIA, May 2026. Joint talk with Eric Kubischta. *Invited seminar.*
- **Icosahedral Quantum Codes from Twisted Unitary t-Groups**, MathQuantum Annual Symposium, University of Maryland, May 2025.
- **Icosahedral Quantum Codes from Twisted Unitary t-Groups**, Quantum Information Processing Conference (QIP 2025), Duke University, February 2025. Joint talk with Eric Kubischta.
- **Symmetry and the Reduction of the Error Correcting Conditions**, Fermilab Quantum Information Science / High Energy Physics Seminar, Fermi National Accelerator Laboratory, July 2023. Joint talk with Eric Kubischta. *Invited seminar.*

Teaching and Mentoring

University of California San Diego — Instructor of Record 2025–present

- MATH 103A, Modern Algebra I — Fall 2025; Winter 2026.
- MATH 18, Linear Algebra — Winter 2026; Fall 2026.
- MATH 102, Applied Linear Algebra — Spring 2026.

University of California San Diego — Undergraduate Mentor 2025–2026

- Guided undergraduate reading and expository projects in lattice-based and isogeny-based cryptography; advised on further coursework and graduate study.

University of Maryland — Instructor of Record 2022

- Linear Algebra MATH 240; Precalculus MATH 113.

University of Maryland — Graduate Student Mentor 2022

- Directed Reading Program in quantum error correction.

University of Maryland — Teaching Assistant 2018–2021

- Linear Algebra MATH 240 (four semesters); Linear Algebra for Scientists and Engineers MATH 461 (two semesters); Calculus III MATH 241; Statistics STAT 100.

Awards and Fellowships

- MathQuantum Fellowship, University of Maryland, 2024.
- Hauptman Summer Fellowship, University of Maryland, 2024.
- Jacob K. Goldhaber Travel Grant, University of Maryland, 2024.
- International Conference Student Support Award, 2024.
- Dean's Fellowship, University of Maryland, 2018–2020.
- Rice Prize, Wesleyan University, 2017 — awarded to the top senior in mathematics.

Professional Service

- Referee for *Physical Review Letters*, *Physical Review X*, *IEEE Transactions on Information Theory*, and *Physical Chemistry Chemical Physics*.